

0613-209: Compiler Design

Theoretical:

1. What is a compiler? Explain the main purpose of a compiler in a programming language environment.
2. Differentiate between a compiler and an interpreter.
3. What are the various phases of a compiler? Briefly explain the function of each phase with a proper figure.
4. What is lexical analysis? What is the role of a lexical analyzer in a compiler?
5. Define the following terms with examples:
 - a. Token
 - b. Lexeme
6. What is syntax analysis (parsing)? Why is it important in the compilation process?
7. What is context-free grammar (CFG)? Explain the different components of a CFG with an example.
8. What is the purpose of a parse tree?
9. Define Ambiguity in CFG with proper examples.

Technical:

1. Given a piece of code and the CFG, divide the program into appropriate lexeme. *Slide-2, Page: 27-32*
2. Write the regular expressions for matching given types of strings. *Slide-2, Page: 14-26*
3. Given a CFG, check if ambiguous. *Slide-3, Page: 18-24*

Syllabus:

1. Slides 1-2
2. Slide 3 (Pages: 1-24)